CHAPTER 2

FEASIBILITY STUDY

2.1 Technical Feasibility

The proposed system is technically feasible using stable and widely supported technologies. The selected technology stack ensures reliability, performance, scalability, and long term maintainability of the application.

The backend of the system is developed using Node.js with Express. The frontend is implemented using React.js to ensure a responsive and dynamic user interface. MongoDB is used as the primary database for storing interview data, transcriptions, insights, and generated documents. Artificial intelligence capabilities are integrated through the Gemini API for transcription, insight extraction, pattern analysis, PRD generation, and chat based editing assistance.

The modular backend design ensures easy debugging and maintenance. AI processing tasks are handled externally through the Gemini API, ensuring that backend performance remains stable. The architecture is scalable and can be deployed on upgraded servers or managed cloud services as future requirements increase.

2.2 Economic Feasibility

The economic feasibility evaluates the financial viability of the system by comparing development and operational costs with the expected benefits. The primary costs include developer time for system development and maintenance, backend and frontend hosting services, optional managed MongoDB hosting, and usage based charges for the Gemini API.

Despite these costs, the system provides significant benefits. It reduces the manual effort required to analyze interviews and generate Product Requirement Documents. Faster processing leads to quicker decision making and improved product quality. The use of open source technologies makes the system cost effective, while AI expenses remain proportional to actual usage.

2.3 Operational Feasibility

The system integrates smoothly with the existing workflow of product managers. The workflow includes interview upload, transcription, insight extraction, PRD generation, and export or ticket creation.

The user interface is intuitive and supports inline editing along with AI assisted chat features. Minimal training is required, and all project data is securely stored in MongoDB for long term retrieval and editing.

2.4 Legal and Ethical Feasibility

The system ensures responsible handling of user data and compliance with relevant privacy standards. Data privacy is maintained through authentication mechanisms, database protection, and secure communication protocols.

Users must provide consent before uploading interview recordings to external AI services. Compliance with the privacy policies and data handling guidelines of the external AI provider is mandatory.

2.5 Schedule Feasibility

The project schedule is realistic and manageable. The core system already demonstrates end to end functionality. The modular architecture allows new features to be added without rewriting the existing codebase, ensuring a predictable and well structured development timeline.